

Finite-Field Rank Tomography and Local–Global Rigidity

for the Alaoglu–Erdős Conjecture

A nonduplicative progress report

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Abstract

Let $x \in \mathbb{R}$ and suppose that 2^x and 3^x are integers. The Alaoglu–Erdős conjecture asserts that x is an integer. The attached unified report already develops a very large collection of analytic, algebraic, determinant, p -adic, interpolation, recurrence, and dynamical approaches. The present report deliberately does not recapitulate that material. Instead it introduces a pair-specific finite-place formalism for a hypothetical counterexample

$$P = (2, 3), \quad Q = (M, A) = (2^\beta, 3^\beta) \in (\mathbb{Q}^\times)^2,$$

where $\beta \notin \mathbb{Z}$ is positive and $M, A \in \mathbb{Z}$.

The main new result is an exact *Smith tomography inequality*. For every good prime power p^t , the number of distinct reductions of a chosen row support under $(a, b) \mapsto P^a Q^b$, and the dual number of character signatures of a chosen column support, bound every layer of the Smith normal form of the associated integer character-evaluation determinant. Short vectors in the resulting congruence lattices yield weighted common-divisor certificates. A discrete-logarithm matrix B_p makes the row and column pictures transposes of one another and gives an exact formula for their full finite-field image sizes. These ingredients lead to a finite, exact, and readily formalizable contradiction certificate: a computable orbit-collapse energy must exceed an explicit archimedean Hadamard budget.

Two local–global consequences are also isolated. Perucca’s cyclic-subgroup detection theorem implies that almost-everywhere local membership $\bar{Q}_p \in \langle \bar{P}_p \rangle$ would already prove the conjecture. A support theorem for reduction orders likewise forces integrality if $\text{ord}(\bar{Q}_p)$ divides $\text{ord}(\bar{P}_p)$ for almost all primes. Conversely, fixed local slopes are rigid, so any useful slopes in a counterexample must drift. A quantitative no-go theorem shows that cyclic image cardinality alone is far too weak; even a weighted certificate built from all short local cyclic slopes cannot close the present Hadamard estimate throughout a substantial range of β . Thus the remaining bridge is sharply identified: one needs many moving short slopes, exceptional p -adic depth, a stronger archimedean determinant bound, or a hybrid of these.

Exact computations for primes below 5000 compare a genuine integral-power pair, two generic pairs, and an archimedean near-miss with logarithmic discrepancy about 1.36×10^{-9} . The integral pair collapses at every tested prime, whereas the near-miss remains finite-field generic at almost every prime. No complete proof of the conjecture is claimed. The report’s contribution is a new exact framework, several rigorous reduction and obstruction theorems, reproducible experiments, and a narrowly specified next proof target.

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1 Scope, status, and nonduplication

1.1 What is and is not being claimed

The status of the central problem is:

The Alaoglu–Erdős conjecture is not proved in this report.

That sentence is not a disclaimer replacing mathematical work. It is a precise boundary around the work below. The report proves new finite-field determinant inequalities, derives exact local–global reductions, identifies quantitative ceilings for tempting strategies, and supplies a reproducible computation. The final implication from the real proportionality

$$(\log M, \log A) = \beta(\log 2, \log 3)$$

to sufficiently abundant finite-place orbit collapse remains open.

Every principal statement is marked conceptually as one of the following.

- **PROVED HERE**: proved in this report.
- **KNOWN / IMPORTED**: a standard theorem or a fact already present in the attached report, imported only as needed.
- **COMPUTATIONAL**: exact finite computation, not a theorem about all primes.
- **OPEN GAP**: a sharply formulated missing bridge or conditional target.

1.2 Nonduplication ledger

The attached manuscript is more than twenty thousand lines long and contains over one hundred sections. It already treats the minimal-counterexample normalization, total positivity of exponential kernels, generalized Vandermonde determinants, universal modular Smith divisors, finite-place and Iwasawa-log methods, gcd bounds, recurrence sequences, interpolation determinants, Kummer theory, Loewner matrices, canonical products, dynamics, entropy, and the restricted Matrix Coefficient Conjecture. Repeating those chapters would obscure rather than advance the problem.

Topic	Treatment here	Relation to the attached report
Counterexample notation	One compact section	Imported baseline only
Exponential determinant	One proposition and one Hadamard bound	Imported baseline only
Universal d -variable Smith divisor	Quoted for scale comparison	Explicitly not claimed as new
Exact row orbit map modulo p^t	Full development	New pair-specific object
Exact dual column-signature map	Full development	New pair-specific object
Smith tomography by orbit counts	Theorem with proof	New
Short relation amplification	Theorem with weighted gcd corollaries	New

Discrete-log image formula and transpose duality	Theorem with proof	New
Full character-table rank criterion	Proposition with proof	New
Perucca support theorems specialized to $(2, 3), (M, A)$	Detailed consequences	New specialization
Cyclic-cardinality ceiling and short-slope ceiling	Asymptotic no-go theorems	New
Prime computation below 5000	Exact reproducible scout	New

1.3 Public-record audit of the claimed rival breakthroughs

Because the prompt explicitly invokes very recent claims, a source audit was performed. This audit is not a mathematical peer review and does not attempt to adjudicate unpublished work.

1. Anthropic publicly released Claude Fable 5 on 11 August 2026 and published a system card describing several research collaborations [2, 3]. Thus the model name is not fictitious.
2. A public preprint by Alpöge gives explicit counterexamples to the Jacobian conjecture in every dimension $n \geq 3$; the public project page describes Fable’s role in the computational search and proof development [4, 5]. This refutes the conjecture in those dimensions; it does not settle the still-distinct plane case $n = 2$.
3. Public project pages report Hadamard constructions including order 668, and one page claims completion of a larger list of previously open orders below 2000 [6, 7]. The public benchmark pages were not fully synchronized when checked, so this report does not certify the exact count stated in the prompt.
4. No citable public construction of an elliptic curve over $\mathbb{Q}$ of algebraic rank at least 30 was located. The public benchmark still described rank 29 as the record and rank 30 as the open target [8]. A private or not-yet-indexed result cannot be ruled out, but it cannot be used as evidence here.
5. No public proof or Lean development of the Alaoglu–Erdős conjecture was located. The rival team’s private message is therefore not independently verifiable.

The appropriate response to competitive pressure is not to lower the standard of proof. It is to extract a mathematically exact target that can be attacked, checked, and formalized.

2 Minimal inherited framework

2.1 Normalization of a hypothetical counterexample

Conjecture 2.1 (Alaoglu–Erdős). *If $x \in \mathbb{R}$ and $2^x, 3^x \in \mathbb{Z}$, then $x \in \mathbb{Z}$.*

Negative x is impossible unless 2^x and 3^x are nonzero integers of absolute value below 1.

Hence any nontrivial counterexample is positive. Assume for the rest of the report that

$$\beta > 0, \quad \beta \notin \mathbb{Z}, \quad M = 2^\beta \in \mathbb{Z}, \quad A = 3^\beta \in \mathbb{Z}.$$

The earlier report proves much stronger finite exclusions; for orientation, one may keep in mind the established lower range $\beta > 27$.

Set

$$T = (\mathbb{G}_m)^2, \quad P = (2, 3), \quad Q = (M, A) \in T(\mathbb{Q}).$$

The real logarithm matrix

$$\Lambda = \begin{pmatrix} \log 2 & \log M \\ \log 3 & \log A \end{pmatrix} = \begin{pmatrix} \log 2 & \beta \log 2 \\ \log 3 & \beta \log 3 \end{pmatrix}$$

has rank one. The finite-place program developed below asks whether this exact real rank-one relation must cast a quantitatively strong “shadow” modulo many prime powers.

Lemma 2.2 (Irrationality and transcendence of β). *A nonintegral β satisfying the displayed hypotheses is irrational and transcendental.*

Proof. If $\beta = a/b \in \mathbb{Q}$ in lowest terms with $b > 0$, then $M^b = 2^a$. Unique factorization forces $b \mid a$, hence $b = 1$ and $\beta \in \mathbb{Z}$, a contradiction. Thus β is irrational. If it were algebraic, then it would be algebraic irrational and the Gelfond–Schneider theorem would make 2^β transcendental, contradicting $M \in \mathbb{Z}$. $\square$

Lemma 2.3 (Global independence). *The points P and Q generate a free rank-two subgroup of $T(\mathbb{Q})$: if $P^u Q^v = (1, 1)$ for $u, v \in \mathbb{Z}$, then $u = v = 0$.*

Proof. The first coordinate gives $2^u M^v = 2^{u+v\beta} = 1$, so $u + v\beta = 0$. Lemma 2.2 makes β irrational, hence $u = v = 0$. $\square$

2.2 The integer character grid

For finite supports

$$\Omega \subset \mathbb{Z}_{\geq 0}^2, \quad \Sigma \subset \mathbb{Z}_{\geq 0}^2,$$

define the character-evaluation matrix

$$E_{\Omega, \Sigma}[(a, b), (r, s)] = \chi_{r, s}(P^a Q^b) = (2^a M^b)^r (3^a A^b)^s. \quad (1)$$

Every entry is an integer. If $|\Omega| = |\Sigma| = m$, write

$$D_{\Omega, \Sigma} = \det E_{\Omega, \Sigma} \in \mathbb{Z}.$$

Using $M = 2^\beta$ and $A = 3^\beta$, the same entry is

$$E_{\Omega, \Sigma}[(a, b), (r, s)] = \exp((a + b\beta)(r \log 2 + s \log 3)). \quad (2)$$

Proposition 2.4 (Nonvanishing baseline; **KNOWN / IMPORTED**). *If the values $a + b\beta$ are distinct on Ω and the values $r \log 2 + s \log 3$ are distinct on Σ , then $D_{\Omega, \Sigma} \neq 0$.*

Proof. After sorting both parameter sets increasingly, the matrix in (2) is a minor of the strictly totally positive kernel $K(x, y) = e^{xy}$. Equivalently, a nonzero exponential polynomial $\sum_{j=1}^m c_j e^{\lambda_j x}$ with distinct real λ_j has at most $m - 1$ real zeros counted with multiplicity; induction after applying $e^{-\lambda_1 x} \frac{d}{dx}$ proves the Chebyshev property. Hence the interpolation matrix is nonsingular. $\square$

For rectangular boxes

$$\Omega_{U,V} = [0, U) \times [0, V), \quad \Sigma_{R,S} = [0, R) \times [0, S), \quad (3)$$

with $UV = RS = m$, distinctness follows from the irrationality of β and of $\log_2 3$. Thus $|D_{\Omega,\Sigma}| \geq 1$. The strategy is to prove that the same nonzero integer is divisible by more prime powers than its archimedean size allows.

3 Exact finite-place orbit maps

3.1 Good primes and the two reduction maps

A prime p is called *good* if

$$p \nmid 6MA.$$

For a good prime and $t \geq 1$, all four coordinates are units modulo p^t . Define the row orbit map

$$\iota_{p^t} : \mathbb{Z}^2 \longrightarrow ((\mathbb{Z}/p^t\mathbb{Z})^\times)^2, \quad (a, b) \longmapsto (2^a M^b, 3^a A^b) \pmod{p^t}. \quad (4)$$

For a finite row support Ω , put

$$c_{p,t}^{\text{row}}(\Omega) = |\iota_{p^t}(\Omega)|. \quad (5)$$

The dual column-signature map is

$$\iota_{p^t}^* : \mathbb{Z}^2 \longrightarrow ((\mathbb{Z}/p^t\mathbb{Z})^\times)^2, \quad (r, s) \longmapsto (2^r 3^s, M^r A^s) \pmod{p^t}, \quad (6)$$

with

$$c_{p,t}^{\text{col}}(\Sigma) = |\iota_{p^t}^*(\Sigma)|. \quad (7)$$

The row map records points of the subgroup generated by P and Q . The column map records how ambient monomial characters restrict to that subgroup.

3.2 Exact congruence lattices

The kernels are rank-two lattices of finite index:

$$L_{p^t}^{\text{row}} = \{(h, k) \in \mathbb{Z}^2 : 2^h M^k \equiv 1, 3^h A^k \equiv 1 \pmod{p^t}\}, \quad (8)$$

$$L_{p^t}^{\text{col}} = \{(r, s) \in \mathbb{Z}^2 : 2^r 3^s \equiv 1, M^r A^s \equiv 1 \pmod{p^t}\}. \quad (9)$$

Consequently,

$$c_{p,t}^{\text{row}}(\Omega) = |\Omega/L_{p^t}^{\text{row}}|, \quad c_{p,t}^{\text{col}}(\Sigma) = |\Sigma/L_{p^t}^{\text{col}}|, \quad (10)$$

where the right-hand side means the number of lattice cosets met by the finite set. This is an exact pair-specific object, not a universal bound obtained merely by counting all residues.

Remark 3.1. The old universal argument only uses

$$c_{p,t}^{\text{row}}(\Omega), c_{p,t}^{\text{col}}(\Sigma) \leq \varphi(p^t)^2 < p^{2t}.$$

The new program computes or bounds the actual box sections of the two lattices. A short lattice vector can make the count dramatically smaller even when the full finite subgroup is large.

4 Pair-specific Smith tomography

4.1 Smith layers as generator counts

Let $E = E_{\Omega, \Sigma}$ be an $m \times m$ integer matrix with nonzero determinant D . Fix a prime p . Its Smith normal form over $\mathbb{Z}_p$ has diagonal entries

$$p^{\nu_1}, \dots, p^{\nu_m}, \quad 0 \leq \nu_1 \leq \dots \leq \nu_m,$$

up to multiplication by p -adic units. For $t \geq 1$ define

$$r_{p,t} = \#\{i : \nu_i < t\}. \quad (11)$$

The elementary Ferrers-diagram identity is

$$v_p(D) = \sum_{i=1}^m \nu_i = \sum_{t \geq 1} (m - r_{p,t}). \quad (12)$$

Only finitely many summands are nonzero.

Lemma 4.1 (Module interpretation). *The integer $r_{p,t}$ is the minimal number of generators of the row module of E over $\mathbb{Z}/p^t\mathbb{Z}$. It is also the minimal number of generators of the column module.*

Proof. Invertible row and column operations do not change either minimal generator number. For the Smith diagonal, the row module is

$$\bigoplus_{\nu_i < t} p^{\nu_i}(\mathbb{Z}/p^t\mathbb{Z}).$$

The ring $\mathbb{Z}/p^t\mathbb{Z}$ is local. Modulo its maximal ideal, each nonzero summand contributes one copy of $\mathbb{F}_p$, so Nakayama's lemma gives exactly one required generator per index with $\nu_i < t$. The column argument is identical. $\square$

4.2 The tomography inequality

Theorem 4.2 (Finite-field Smith tomography; **PROVED HERE**). *Let p be good, let $t \geq 1$, and let $E_{\Omega, \Sigma}$ be square of size m with nonzero determinant. Then*

$$r_{p,t} \leq \min(c_{p,t}^{\text{row}}(\Omega), c_{p,t}^{\text{col}}(\Sigma)). \quad (13)$$

Consequently,

$$v_p(D_{\Omega, \Sigma}) \geq \sum_{t \geq 1} \left(m - \min(c_{p,t}^{\text{row}}(\Omega), c_{p,t}^{\text{col}}(\Sigma)) \right). \quad (14)$$

Proof. Reduce the matrix modulo p^t . If two row indices $(a, b), (a', b')$ have the same image under ι_{p^t} , then

$$(2^a M^b, 3^a A^b) \equiv (2^{a'} M^{b'}, 3^{a'} A^{b'}) \pmod{p^t}.$$

Every monomial character therefore takes the same value on those two points, so the corresponding matrix rows are identical modulo p^t . The row module is generated by at most one representative from each orbit class met by Ω ; hence its minimal number of generators is at most $c_{p,t}^{\text{row}}(\Omega)$. Lemma 4.1 gives $r_{p,t} \leq c_{p,t}^{\text{row}}(\Omega)$.

Likewise, if two columns have the same image under $\iota_{p^t}^*$, then their quotient character is trivial on both generators P and Q , hence on every row point $P^a Q^b$. The two columns are identical modulo p^t , and $r_{p,t} \leq c_{p,t}^{\text{col}}(\Sigma)$. This proves (13). Summing $m - r_{p,t}$ and using (12) proves (14). $\square$

Remark 4.3 (Why this is stronger than rank modulo p). The ordinary rank defect modulo p controls only $m - r_{p,1}$. Theorem 4.2 controls every threshold t and therefore every layer of the Smith profile. A congruence persisting from p to p^t is counted t times, exactly as its valuation demands.

Corollary 4.4 (Global orbit-collapse divisor). *For any finite set $\mathcal{P}$ of good primes,*

$$\prod_{p \in \mathcal{P}} p^{\sum_{t \geq 1} (m - \min(c_{p,t}^{\text{row}}(\Omega), c_{p,t}^{\text{col}}(\Sigma)))} \mid D_{\Omega, \Sigma}. \quad (15)$$

Proof. Apply Theorem 4.2 prime by prime. Distinct prime powers are coprime. $\square$

4.3 Recovery of the universal bound and the genuinely new term

The old universal estimate is recovered by the trivial inequalities

$$c_{p,t}^{\text{row}}, c_{p,t}^{\text{col}} \leq p^{2t}.$$

Thus one gets

$$v_p(D) \geq \sum_{t \geq 1} (m - p^{2t})_+,$$

and, after summing over primes,

$$\log |D| \geq \left(\frac{2}{3} + o(1) \right) m^{3/2}.$$

This 2/3 law is already in the attached report and is quoted only for comparison. The new information is the nonnegative excess

$$\Delta_{p,t} = p^{2t} - \min(c_{p,t}^{\text{row}}, c_{p,t}^{\text{col}}), \quad (16)$$

whenever the actual pair occupies fewer residues than the ambient two-dimensional torus permits. The central quantitative problem is whether the sum of these pair-specific excesses can be forced to reach the archimedean scale m^2 .

5 Short congruence relations and weighted divisors

5.1 A box-translation graph

Let $\Omega_{U,V}$ be the row box in (3). For a nonzero vector $u = (h, k) \in \mathbb{Z}^2$ with $|h| < U$ and $|k| < V$, join z to $z + u$ whenever both vertices lie in the box. Orient each edge in the u direction after replacing u by $-u$ if necessary.

Lemma 5.1 (Translation forest). *The graph is a disjoint union of paths and has exactly*

$$e_u = (U - |h|)(V - |k|) \quad (17)$$

edges. Hence it has $UV - e_u$ connected components.

Proof. The overlap of the box with its translate by $-u$ has the stated cardinality, giving the edge count. A directed cycle would imply $nu = 0$ in $\mathbb{Z}^2$ for some $n > 0$, impossible because $u \neq 0$. Each vertex has at most one outgoing and one incoming edge, so every component is a path. A forest with UV vertices and e_u edges has $UV - e_u$ components. $\square$

5.2 Short relation amplification

Theorem 5.2 (Short row relation amplification; **PROVED HERE**). *Let p be good and suppose that*

$$P^h Q^k \equiv (1, 1) \pmod{p^t} \quad (18)$$

for a nonzero $(h, k) \in \mathbb{Z}^2$ with $|h| < U$ and $|k| < V$. For any column support Σ of size UV for which the evaluation matrix is square and nonsingular,

$$v_p(D_{\Omega_{U,V},\Sigma}) \geq t(U - |h|)(V - |k|). \quad (19)$$

The analogous assertion holds with rows and columns interchanged and a short vector in $L_{p^t}^{\text{col}}$.

Proof. Along every edge $z \rightarrow z + (h, k)$ of the translation forest, relation (18) makes the two row points congruent modulo p^t , hence their matrix rows are congruent. The row orbit map is constant on every connected component. Lemma 5.1 gives

$$c_{p,j}^{\text{row}}(\Omega_{U,V}) \leq UV - (U - |h|)(V - |k|)$$

for every $1 \leq j \leq t$. Apply the row half of Theorem 4.2 at each level j and sum. $\square$

The result is stronger than saying that the determinant vanishes modulo p^t . A single relation is replicated across every overlapping translate in the box, and each replication contributes an independent Smith defect.

5.3 Weighted gcd certificates

For $0 \leq h < U$ and $0 \leq k < V$, not both zero, define

$$G_{h,k}^{\text{row}} = \gcd(2^h M^k - 1, 3^h A^k - 1). \quad (20)$$

For $0 \leq r < R$, $0 \leq s < S$, not both zero, define

$$G_{r,s}^{\text{col}} = \gcd(2^r 3^s - 1, M^r A^s - 1). \quad (21)$$

Corollary 5.3 (Weighted common divisors; **PROVED HERE**). *For a square nonsingular box matrix with $UV = RS = m$,*

$$\left(G_{h,k}^{\text{row}}\right)^{(U-h)(V-k)} \mid D, \quad (22)$$

$$\left(G_{r,s}^{\text{col}}\right)^{(R-r)(S-s)} \mid D. \quad (23)$$

Consequently, the least common multiple of all prime-power-weighted quantities on either line divides D . A product is not generally valid because the same prime valuation may be counted by several relations.

Proof. If p^t divides $G_{h,k}^{\text{row}}$, then $P^h Q^k \equiv 1 \pmod{p^t}$. The direct row-difference argument in Theorem 5.2 works even at the finitely many primes excluded from the group notation, because the two displayed congruences themselves are sufficient. Thus

$$v_p(D) \geq t(U - h)(V - k).$$

Take $t = v_p(G_{h,k}^{\text{row}})$ for every prime. The column case is identical. $\square$

Remark 5.4 (Why the lcm matters). The attached report correctly emphasizes that multiplying many gcd divisors can overcount a prime. The exact safe aggregate is

$$\text{lcm}_{h,k} \left(G_{h,k}^{\text{row}} \right)^{(U-h)(V-k)},$$

whose p -valuation is the maximum, not the sum, of the competing weighted valuations. The full Smith tomography energy can be larger because several independent lattice directions may create more components than any single relation detects.

6 Discrete logarithms and transpose duality

6.1 The local orbit matrix

Let p be an odd good prime, put $n = p - 1$, choose a primitive root g modulo p , and write $\ell_p(u) \in \mathbb{Z}/n\mathbb{Z}$ for the discrete logarithm of u . Define

$$B_p = \begin{pmatrix} \ell_p(2) & \ell_p(M) \\ \ell_p(3) & \ell_p(A) \end{pmatrix} \pmod{n}. \quad (24)$$

Under the identification $(\mathbb{F}_p^\times)^2 \cong (\mathbb{Z}/n\mathbb{Z})^2$, the row map is multiplication by B_p :

$$(a, b)^T \mapsto B_p(a, b)^T.$$

The column-signature map is multiplication by the transpose:

$$(r, s)^T \mapsto B_p^T(r, s)^T.$$

This simple transpose is the finite-field origin of the row–column symmetry in Theorem 4.2.

Theorem 6.1 (Exact image size; **PROVED HERE**). *Choose arbitrary integer lifts b_{ij} of the entries of B_p and set*

$$q_p = \gcd(n^2, nb_{11}, nb_{12}, nb_{21}, nb_{22}, \det B_p). \quad (25)$$

Then

$$|\text{im } B_p| = |\text{im } B_p^T| = \frac{n^2}{q_p}. \quad (26)$$

The formula is independent of the chosen primitive root and of the integer lifts.

Proof. Let $L \subset \mathbb{Z}^2$ be the lattice generated by the columns of the 2×4 matrix

$$[nI_2 \mid B_p].$$

The quotient $\mathbb{Z}^2/L$ is the cokernel of $B_p : (\mathbb{Z}/n\mathbb{Z})^2 \rightarrow (\mathbb{Z}/n\mathbb{Z})^2$. Its order is the index $[\mathbb{Z}^2 : L]$. For a full-rank integer matrix with two rows, that index is the gcd of its 2×2 minors. Those minors are precisely the six integers in (25), up to signs. Therefore the cokernel has order q_p , and the image has order n^2/q_p . Transposition leaves the list of entries and the determinant unchanged up to sign, so it has the same image size. Since the result computes an intrinsic group cardinality, it is independent of all choices. $\square$

Corollary 6.2 (Rank-one-scale reduction). *If*

$$\det B_p \equiv 0 \pmod{p-1}, \quad (27)$$

then

$$|\operatorname{im} B_p| = |\operatorname{im} B_p^T| \leq p-1.$$

Proof. Every number in the gcd (25) is then divisible by $n = p-1$, so $q_p \geq n$. $\square$

The terminology “rank-one-scale” is deliberate. Over the composite ring $\mathbb{Z}/(p-1)\mathbb{Z}$, determinant zero need not mean that the image is a free cyclic module, but its cardinality is at most the one-dimensional ambient scale $p-1$. If $Q = P^c$ globally, then the second column of B_p is c times the first for every good prime, so (27) holds everywhere.

6.2 Exact rank when full local groups are sampled

Let $H_p = \langle \bar{P}_p, \bar{Q}_p \rangle \subset (\mathbb{F}_p^\times)^2$. The ambient monomial characters restrict to characters of H_p .

Proposition 6.3 (Full character-table rank; **PROVED HERE**). *Suppose the row support maps onto all of H_p and the column support maps onto all distinct restricted characters of H_p . Then the reduction of $E_{\Omega, \Sigma}$ modulo p has rank exactly*

$$|H_p| = |\operatorname{im} B_p|.$$

Proof. The image of B_p^T has the same size as H_p by Theorem 6.1; it is exactly the set of restriction signatures of ambient characters. Choose one row representative for each $h \in H_p$ and one column representative for each restricted character $\chi \in \hat{H}_p$. The resulting square matrix is the character table $(\chi(h))$. Its rows are orthogonal: for a nontrivial character ψ , choosing h_0 with $\psi(h_0) \neq 1$ gives

$$\sum_{h \in H_p} \psi(h) = \sum_{h \in H_p} \psi(h_0 h) = \psi(h_0) \sum_{h \in H_p} \psi(h),$$

so the sum vanishes. Since $p \nmid |H_p|$, the character table is invertible over $\mathbb{F}_p$. Duplicating rows or columns cannot increase rank, proving the claim. $\square$

Remark 6.4. Proposition 6.3 distinguishes two causes of small determinant rank: a genuinely small subgroup H_p , and failure of the finite boxes to cover the available group or dual group. For the determinant method, the second phenomenon is often more important. A huge cyclic subgroup can still produce many box collisions when its local slope is short.

7 Local slopes: rigidity, divisibility, and the need to drift

The congruence lattice becomes especially concrete when the reduction of Q lies in the cyclic subgroup generated by the reduction of P . This is the finite-place analogue of a rational slope. It is important to distinguish three assertions:

- (a) the subgroup $\langle \bar{P}_p, \bar{Q}_p \rangle$ has small cardinality;
- (b) $\bar{Q}_p$ lies in $\langle \bar{P}_p \rangle$;
- (c) $\bar{Q}_p = \bar{P}_p^c$ for a *small* integer representative c .

The determinant sees increasingly more information as one passes from (a) to (c). A cyclic image can still have order about p ; local membership can require a slope of size about p ; only a short slope gives a large collision inside a bounded exponent box.

7.1 Exact counting from one local slope

Let G be a finite group, let $P_0 \in G$ have order d , and suppose $Q_0 = P_0^c$. For integers $U, V \geq 1$ consider

$$\psi_{U,V}(a, b) = P_0^a Q_0^b = P_0^{a+cb}, \quad 0 \leq a < U, \quad 0 \leq b < V.$$

Proposition 7.1 (Local-slope image bound). *Choose an integer representative c of the local slope. Then*

$$\#\psi_{U,V}([0, U) \times [0, V)) \leq \min\{d, U + |c|(V - 1)\}.$$

If $0 \leq c < U$, the relation $(-c, 1)$ produces at least

$$(U - c)(V - 1)$$

row collisions in the sense of Theorem 5.2.

Proof. The integers $a + cb$ lie in an interval of length at most $U - 1 + |c|(V - 1)$, so they assume at most $U + |c|(V - 1)$ values before reduction modulo d , and never more than d values after reduction. If $0 \leq c < U$, then

$$P_0^{-c} Q_0 = 1.$$

The two boxes $[0, U) \times [0, V)$ and its translate by $(-c, 1)$ overlap in exactly $(U - c)(V - 1)$ points. Apply Theorem 5.2. $\square$

For the Alaoglu–Erdős pair this says that a congruence

$$(M, A) \equiv (2^c, 3^c) \pmod{p^t}$$

with $0 \leq c < U$ contributes the factor

$$p^{t(U-c)(V-1)}$$

to the row-grid determinant. The same assertion can be read directly from a two-coordinate gcd.

Definition 7.2 (Slope gcd). For $c \in \mathbb{N}_0$ put

$$H_c := \gcd(|M - 2^c|, |A - 3^c|).$$

For a nonintegral counterexample, H_c is always a positive integer: $M = 2^c$ and $A = 3^c$ cannot occur simultaneously.

Corollary 7.3 (Weighted local-slope divisor). *Let D be the determinant associated with the row box $[0, U) \times [0, V)$ and any equally large admissible column support. For every $0 \leq c < U$,*

$$H_c^{(U-c)(V-1)} \mid D.$$

Consequently

$$\mathfrak{C}_{U,V} := \text{lcm}_{0 \leq c < U} H_c^{(U-c)(V-1)} \quad \text{satisfies} \quad \mathfrak{C}_{U,V} \mid D. \quad (28)$$

Proof. If $p^t \mid H_c$, then $Q \equiv P^c \pmod{p^t}$. Proposition 7.1 and Theorem 5.2 give $v_p(D) \geq t(U - c)(V - 1)$. This holds prime by prime. Taking the least common multiple is the exact way to combine possibly overlapping prime factors. $\square$

7.2 A fixed slope cannot recur indefinitely

A key rigidity point is elementary but decisive. Any local slopes capable of contributing at unbounded primes must move.

Theorem 7.4 (Fixed-slope rigidity). *Fix $c \in \mathbb{Z}$. If*

$$\bar{Q}_p = \bar{P}_p^c$$

for infinitely many good primes p , then $Q = P^c$ in $T(\mathbb{Q})$. In particular, under a nonintegral counterexample every fixed slope occurs at only finitely many good primes.

Proof. For $c \geq 0$, every such prime divides both $M - 2^c$ and $A - 3^c$. Infinitely many prime divisors force these two fixed integers to vanish, so $M = 2^c$ and $A = 3^c$. For $c < 0$, multiply the congruences by 2^{-c} and 3^{-c} : every such prime divides both $2^{-c}M - 1$ and $3^{-c}A - 1$, and the same argument applies. The conclusion $Q = P^c$ makes $\beta = c$. $\square$

The proof also gives an exact weighted version. For $c \geq 0$,

$$\sum_{\substack{p \text{ good} \\ \bar{Q}_p = \bar{P}_p^c}} \log p \leq \log \text{rad } H_c. \quad (29)$$

Thus a successful finite-field argument cannot repeatedly reuse a bounded collection of slopes. It must extract a large logarithmic weight from slopes c_p that drift with p , or from high powers p^t dividing the corresponding H_{c_p} .

7.3 An intrinsic ceiling for the all-short-slope lcm

The divisor (28) is exact, but its maximum possible size can be bounded without any transcendence theory. This produces a useful no-go theorem for the most naive short-slope strategy.

Proposition 7.5 (Archimedean ceiling for the slope certificate). *For fixed M, A and $U, V \rightarrow \infty$,*

$$\begin{aligned} \log \mathfrak{C}_{U,V} &\leq (V-1) \sum_{c=0}^{U-1} (U-c) \log^+ H_c \\ &\leq (V-1) \sum_{c=0}^{U-1} (U-c) \log^+ |M - 2^c| \\ &= \frac{\log 2}{6} VU^3 + O_M(VU^2). \end{aligned} \quad (30)$$

Here $\log^+ y = \max(0, \log y)$.

Proof. The logarithm of a least common multiple is bounded by the logarithm of the corresponding product, which gives the first line. Since $H_c \leq |M - 2^c|$ unless the latter vanishes, and simultaneous vanishing is excluded, the second line follows after harmless treatment of finitely many small c . Finally

$$\log |M - 2^c| = c \log 2 + O_M(2^{-c})$$

for large c , while

$$\sum_{c=0}^{U-1} (U-c)c = \frac{U(U-1)(U+1)}{6}.$$

□

Suppose the row box is aspect-optimized with area $m = UV$ and $U^2 \sim \beta m$, as in Section 8. Then Proposition 7.5 becomes

$$\log \mathfrak{C}_{U,V} \leq \left(\frac{\beta \log 2}{6} + o(1) \right) m^2. \quad (31)$$

The optimized Hadamard budget below has leading constant $4\sqrt{\beta \log 2 \log 3}$. Hence the *maximum possible upper envelope* of this particular lcm certificate is smaller than that Hadamard constant whenever

$$\beta < 576 \log_2 3 = 912.96 \dots \quad (32)$$

This comparison does not upper-bound the true p -adic valuation of the determinant; independent relations and higher Smith layers can contribute more. It does prove that merely listing all congruences $Q \equiv P^c$ with $0 \leq c < U$ and combining their individual gcds by an lcm cannot, by itself, close the present Hadamard estimate throughout the range (32).

Remark 7.6 (Why this matters). The finite computation in Section 11 finds exactly the behavior predicted by Theorem 7.4. For the genuine integral pair $Q = P^3$, the same slope appears at every good prime. For an extremely close real near-miss, the few local slopes that occur are isolated and unrelated. Archimedean closeness alone does not create a stable finite-field slope.

8 The finite contradiction criterion and its quantitative scale

The preceding sections supply a lower bound for a positive integer determinant. To turn it into a contradiction one needs an upper bound for the same determinant. The deliberately coarse Hadamard estimate is enough to formulate an exact finite certificate and to expose the missing scale.

8.1 An explicit archimedean budget

Let

$$\Omega_{U,V} = \{0, \dots, U-1\} \times \{0, \dots, V-1\}, \quad \Sigma_{R,S} = \{0, \dots, R-1\} \times \{0, \dots, S-1\},$$

and suppose

$$m := UV = RS.$$

Write $D_{U,V;R,S}$ for the determinant of the corresponding square evaluation matrix. Under a hypothetical nonintegral solution, total positivity gives $D_{U,V;R,S} > 0$.

Set

$$T_{U,V} := (U-1) + \beta(V-1), \quad L_{R,S} := (R-1) \log 2 + (S-1) \log 3. \quad (33)$$

Every entry has the form $e^{t\lambda}$ with $0 \leq t \leq T_{U,V}$ and $0 \leq \lambda \leq L_{R,S}$.

Theorem 8.1 (Hadamard budget). *One has*

$$0 < \log D_{U,V;R,S} \leq m T_{U,V} L_{R,S} + \frac{m}{2} \log m. \quad (34)$$

Proof. Each row contains m positive entries, all at most $e^{T_{U,V} L_{R,S}}$. Its Euclidean norm is therefore at most $\sqrt{m} e^{T_{U,V} L_{R,S}}$. Hadamard’s inequality gives

$$D_{U,V;R,S} \leq \left(\sqrt{m} e^{T_{U,V} L_{R,S}} \right)^m.$$

The determinant is positive by the exponential-kernel argument in Section 2. $\square$

8.2 Exact orbit-collapse energy

For every good prime power define

$$c_{p,t}^{\text{row}}(U, V) := \#\iota_{p^t}(\Omega_{U,V}), \quad c_{p,t}^{\text{col}}(R, S) := \#\iota_{p^t}^*(\Sigma_{R,S}).$$

The associated energy is

$$\mathcal{A}_{U,V;R,S} := \sum_{p \nmid 6MA} \sum_{t \geq 1} \left(m - \min\{c_{p,t}^{\text{row}}(U, V), c_{p,t}^{\text{col}}(R, S)\} \right) \log p. \quad (35)$$

Only finitely many summands are nonzero for fixed boxes: once the reductions of all rows and columns are pairwise distinct modulo p^t , the corresponding term vanishes.

Theorem 8.2 (Finite orbit-collapse certificate). *If, for some positive integers U, V, R, S with $UV = RS = m$,*

$$\mathcal{A}_{U,V;R,S} > m T_{U,V} L_{R,S} + \frac{m}{2} \log m, \quad (36)$$

then no nonintegral Alaoglu–Erdős solution can have output pair (M, A) .

Proof. The Smith tomography theorem gives

$$\log D_{U,V;R,S} \geq \mathcal{A}_{U,V;R,S},$$

whereas Theorem 8.1 gives the opposite upper bound. Strict inequality (36) is impossible. $\square$

This is a proof-producing criterion. It involves only exact modular exponentiation, counting finite images, and a rigorously bounded real expression. A finite list of primes and levels already gives a valid lower bound by truncating (35); no unproved distribution statement is needed to certify any successful instance.

8.3 Optimizing the rectangle shapes

The row and column boxes should not generally be square. For fixed area m , the leading parts of $T_{U,V}$ and $L_{R,S}$ are minimized separately by balancing their two summands:

$$U \sim \sqrt{\beta m}, \quad V \sim \sqrt{m/\beta}, \quad (37)$$

$$R \sim \sqrt{m \log_2 3}, \quad S \sim \sqrt{m \log_3 2}. \quad (38)$$

Indeed, the arithmetic–geometric mean inequality gives

$$\begin{aligned} U + \beta V &\geq 2\sqrt{\beta UV} = 2\sqrt{\beta m}, \\ R \log 2 + S \log 3 &\geq 2\sqrt{RS \log 2 \log 3} = 2\sqrt{m \log 2 \log 3}. \end{aligned} \quad (39)$$

After rounding the four side lengths through a sequence of admissible factorizations of m ,

$$mT_{U,V}L_{R,S} = \left(4\sqrt{\beta \log 2 \log 3} + o(1)\right) m^2. \quad (40)$$

The optimization reduces the large- β dependence from linear to square-root, but the scale remains m^2 .

8.4 Why cyclic image size alone is not enough

Suppose, unrealistically favorably, that for every good prime power the full row image were cyclic. Its size would then be at most $\varphi(p^t)$. Replacing the exact orbit count by this bound yields the formal lower certificate

$$\mathcal{C}(m) := \sum_p \sum_{t \geq 1} (m - \varphi(p^t))_+ \log p. \quad (41)$$

Proposition 8.3 (Cyclic-cardinality ceiling). *As $m \rightarrow \infty$,*

$$\mathcal{C}(m) = \frac{1}{2}m^2 + o(m^2). \quad (42)$$

Proof. The level $t = 1$ contributes

$$\sum_{p \leq m+1} (m - p + 1) \log p = m\vartheta(m) - \sum_{p \leq m} p \log p + o(m^2),$$

where ϑ is Chebyshev’s function. Partial summation and the prime number theorem give $\vartheta(m) \sim m$ and $\sum_{p \leq m} p \log p \sim m^2/2$, hence the main term $m^2/2$. For $t \geq 2$, the condition $\varphi(p^t) < m$ forces $p \ll m^{1/t}$; summing the trivial bound $m \log p$ gives $O(m^{3/2})$ at $t = 2$ and smaller orders thereafter. $\square$

Compare the constant $1/2$ with the optimized Hadamard constant $4\sqrt{\beta \log 2 \log 3}$. Even for $\beta = 1$, the latter exceeds 3.49. Thus *cyclic reduction image at every finite level* is quantitatively insufficient under the present upper bound. The determinant must see more than local rank one: it needs short slopes, repeated high-power congruence, multiple independent relations, or a significantly sharper archimedean estimate.

8.5 A quantitative moving-slope target

The exact short-relation theorem turns the preceding diagnosis into a clean sufficient condition. Fix $0 < \eta < 1$ and let $\mathcal{P}_m$ be a set of good primes for which there exists an integer c_p satisfying

$$|c_p| \leq \eta U, \quad \bar{Q}_p = \bar{P}_p^{c_p}.$$

Choose the sign convention so that the overlap of the row box with its translate by $(-c_p, 1)$ is at least $(1 - \eta)U(V - 1)$. Then

$$\mathcal{A}_{U,V;R,S} \geq (1 - \eta)U(V - 1) \sum_{p \in \mathcal{P}_m} \log p. \quad (43)$$

Since $U(V-1) = m + O(U)$, an aspect-optimized sequence would contradict the hypothetical pair if

$$\sum_{p \in \mathcal{P}_m} \log p > \left(\frac{4\sqrt{\beta \log 2 \log 3}}{1 - \eta} + o(1) \right) m. \quad (44)$$

A p^t congruence receives the weight $t \log p$, so the same criterion admits depth in place of more primes.

Condition (44) is the first genuinely quantitative target of this report. It asks for logarithmic prime weight of order m , not density among all primes, and it explicitly allows the slopes to drift as Theorem 7.4 requires. What remains unknown is a theorem transferring the singular real logarithm matrix into this much finite-place short-slope mass.

9 Local–global support theorems specialized to the conjecture

The finite-field framework naturally meets the classical support problem for algebraic groups. The relevant results are due to Corrales–Rodrigáñez and Schoof, and in the general product-of-tori-and-abelian-varieties form to Perucca [9, 10, 11, 12]. Their qualitative conclusions are strong enough to give exact conditional proofs of Alaoglu–Erdős under almost-everywhere local hypotheses. They do not supply the quantitative short relations required by (44).

9.1 Cyclic subgroup detection

We use the following imported theorem in its torus specialization.

Theorem 9.1 (Perucca, cyclic reduction detection). *Let G be a product of an abelian variety and a torus over a number field, let $R \in G(K)$, and let $\Lambda \subset G(K)$ be a free cyclic subgroup. If the reduction of R belongs to the reduction of Λ for all but finitely many primes of good reduction, then $R \in \Lambda$.*

Apply this with $G = T = (\mathbb{G}_m)^2$, $R = Q$, and $\Lambda = \langle P \rangle$.

Corollary 9.2 (Almost-everywhere local cyclicity proves Alaoglu–Erdős). *If*

$$\bar{Q}_p \in \langle \bar{P}_p \rangle$$

for all but finitely many good primes p , then $\beta \in \mathbb{Z}$.

Proof. Theorem 9.1 gives $Q = P^n$ for some $n \in \mathbb{Z}$. Positivity and $M, A > 1$ give $n > 0$, and then $M = 2^n$, $A = 3^n$, so $\beta = n$. $\square$

The contrapositive is unconditional and useful:

$$\beta \notin \mathbb{Z} \implies \bar{Q}_p \notin \langle \bar{P}_p \rangle \text{ for infinitely many good primes } p. \quad (45)$$

It says that a counterexample cannot be locally rank one almost everywhere. Qualitative support theory therefore creates infinitely many *anti-collapse witnesses*. A proof via determinant collapse must coexist with these witnesses and concentrate on a sufficiently large complementary set.

9.2 Order support already suffices

A second support theorem uses only the orders of the reductions. In a torus, endomorphisms are integral matrices.

Theorem 9.3 (Perucca, order support; torus form). *Let $P, Q \in T(\mathbb{Q})$. If*

$$\text{ord}(\bar{Q}_p) \mid \text{ord}(\bar{P}_p)$$

for all but finitely many good primes, then there exist a nonzero integer c and an endomorphism $\phi \in \text{End}_{\mathbb{Q}}(T) \cong M_2(\mathbb{Z})$ such that

$$\phi(P) = Q^c.$$

Corollary 9.4 (Almost-everywhere order divisibility proves the conjecture). *Under the hypothetical output relation $M = 2^\beta$, $A = 3^\beta$, if*

$$\text{ord}(\bar{Q}_p) \mid \text{ord}(\bar{P}_p)$$

for all but finitely many good primes, then $\beta \in \mathbb{Z}$.

Proof. Write

$$\phi = \begin{pmatrix} u & v \\ r & s \end{pmatrix}.$$

The equation $\phi(P) = Q^c$ says

$$M^c = 2^u 3^v, \quad A^c = 2^r 3^s.$$

Every prime divisor of M or A is therefore 2 or 3, so write

$$M = 2^{u_0} 3^{v_0}, \quad A = 2^{r_0} 3^{s_0}$$

with nonnegative exponents. Put $\theta = \log_2 3$. The equality $2^\beta = M$ gives $\beta = u_0 + v_0\theta$, while $3^\beta = A$ gives, after taking logarithms base 2,

$$\beta\theta = r_0 + s_0\theta.$$

Eliminating β yields

$$v_0\theta^2 + (u_0 - s_0)\theta - r_0 = 0.$$

The number θ is transcendental: it is irrational by unique factorization, and if it were algebraic then the Gelfond–Schneider theorem applied to $2^\theta = 3$ would be contradicted. Hence every coefficient in the displayed polynomial is zero. Thus $v_0 = r_0 = 0$ and $u_0 = s_0$, so $M = 2^{u_0}$, $A = 3^{u_0}$ and $\beta = u_0$. $\square$

Again the contrapositive supplies an infinite local signature:

$$\beta \notin \mathbb{Z} \implies \text{ord}(\bar{Q}_p) \nmid \text{ord}(\bar{P}_p) \text{ for infinitely many good } p. \quad (46)$$

Perucca also proves variants that compare the ℓ -adic valuations of the two orders for a fixed auxiliary prime ℓ [10]. Those variants are particularly relevant to the shape of the congruence lattice.

9.3 Order imbalance forces long coefficients

Lemma 9.5 (Order-imbalance obstruction). *Let G be a finite abelian group, $P_0, Q_0 \in G$, and suppose*

$$P_0^a Q_0^b = 1.$$

For every prime ℓ ,

$$v_\ell(b) \geq (v_\ell(\text{ord } Q_0) - v_\ell(\text{ord } P_0))_+. \quad (47)$$

The symmetric assertion holds with a and b interchanged.

Proof. The relation gives $Q_0^b = P_0^{-a}$, whose order divides $\text{ord } P_0$. Since

$$v_\ell(\text{ord}(Q_0^b)) = \max\{v_\ell(\text{ord } Q_0) - v_\ell(b), 0\},$$

comparison with $v_\ell(\text{ord } P_0)$ gives (47). $\square$

This lemma shows why support-theorem witness primes need not help the determinant. If the ℓ -part of $\text{ord}(\bar{Q}_p)$ exceeds that of $\text{ord}(\bar{P}_p)$ by j , then every local relation with nonzero Q -coefficient has that coefficient divisible by ℓ^j . For a box with $V \leq \ell^j$, no such relation is short in the vertical direction. The same primes that certify global nondependence can therefore make the reduction lattice *less* compressible.

Qualitative local–global theory reaches an exact logical endpoint:

- local cyclicity almost everywhere would prove $\beta \in \mathbb{Z}$;
- local order domination almost everywhere would prove $\beta \in \mathbb{Z}$;
- a counterexample must have infinitely many noncyclic and order-imbalance witness primes;
- none of these statements controls the logarithmic mass of primes carrying *short* relations.

The missing theorem is quantitative and geometric in the two-dimensional relation lattice, not merely qualitative membership in a subgroup.

10 Fixed-ray common divisors and why moving directions are essential

The weighted gcd divisors of Section 5 might suggest taking one promising relation direction and scaling it. The theorem of Bugeaud, Corvaja, and Zannier rules out that architecture at exponential scale [14].

10.1 The row rays

Fix $(h, k) \in \mathbb{N}_0^2 \setminus \{(0, 0)\}$ and put

$$U_0 = 2^h M^k, \quad V_0 = 3^h A^k.$$

Under a hypothetical counterexample,

$$\log U_0 = (h + k\beta) \log 2, \quad \log V_0 = (h + k\beta) \log 3.$$

The integers U_0 and V_0 are multiplicatively independent. Indeed, a relation $U_0^r = V_0^s$ would imply $r \log 2 = s \log 3$ because $h + k\beta > 0$, which is impossible unless $r = s = 0$.

The Bugeaud–Corvaja–Zannier theorem therefore gives, for every $\varepsilon > 0$ and all sufficiently large n ,

$$\log \gcd(U_0^n - 1, V_0^n - 1) < \varepsilon n. \quad (48)$$

But this gcd is exactly

$$G_{nh,nk}^{\text{row}}.$$

Thus no fixed row direction carries a common divisor of positive exponential rate.

10.2 The column rays

Fix $(r, s) \in \mathbb{N}_0^2 \setminus \{(0, 0)\}$ and put

$$W_0 = 2^r 3^s, \quad Z_0 = M^r A^s = W_0^\beta.$$

These are multiplicatively independent: if $W_0^u = Z_0^v$, then $W_0^{u-v\beta} = 1$, so $u = v\beta$; irrationality of β forces $u = v = 0$. Therefore

$$\log \gcd(W_0^n - 1, Z_0^n - 1) < \varepsilon n \quad (49)$$

for large n , and the gcd is $G_{nr,ns}^{\text{col}}$.

10.3 A finite-family no-go theorem

Proposition 10.1 (Finitely many fixed rays are lower order). *Fix finitely many nonzero row and column directions. Let the determinant boxes have side lengths $O(N)$ in each coordinate, and combine the weighted gcd divisors from all integer multiples of those directions that fit in the boxes. The logarithm of the resulting lcm is $o(N^4)$. In particular it is $o(m^2)$ when the matrix size is $m \asymp N^2$.*

Proof. For one fixed direction, a multiple n that remains inside the box satisfies $n = O(N)$, and its overlap weight is $O(N^2)$. By (48) or (49), if g_n denotes the associated gcd, then $\log g_n = o(n)$. Hence

$$\sum_{n \leq CN} \log g_n = o(N^2).$$

The logarithm of the lcm of the weighted divisors is at most the logarithm of their product, and therefore at most $O(N^2) \sum_{n \leq CN} \log g_n = o(N^4)$. A fixed finite sum over directions remains $o(N^4)$. $\square$

The conclusion is stronger than the statement that one particular gcd is small. It says that any closing common-divisor mechanism must use a number of relation directions growing with the determinant, and those directions must move through the lattice. This matches independently the fixed-slope rigidity of Theorem 7.4 and the conditional target (44).

Remark 10.2 (Uniformity is the open issue). The Bugeaud–Corvaja–Zannier bound is asymptotic for fixed U_0, V_0 . The bases in the moving-direction problem themselves grow with the box. A uniform theorem over two-parameter families, with information about the *union* of prime divisors rather than a single gcd, would be a genuinely new input. Existing S -unit and Subspace-Theorem technology does not provide the needed lower or upper control at this moving scale.

11 Exact finite-field reconnaissance

Theorems 4.2 and 8.2 are exact, but they leave open how exceptional the orbit collapse is for actual integer pairs. I wrote a dependency-free Python scout to measure the one-level row image

$$\{(2^a M^b, 3^a A^b) \bmod p : 0 \leq a, b < N\}$$

for every good prime $p \leq 5000$. The purpose was diagnostic, not evidentiary: the calculation does not assume the conjecture, and no numerical observation is promoted to a theorem.

11.1 Protocol

The run used $N = 24$, hence $m = N^2 = 576$. For each pair (M, A) and each good prime, the program computed:

- (i) the exact row-orbit image size c_p and defect $m - c_p$;
- (ii) the orders of $P = (2, 3)$ and $Q = (M, A)$ in $(\mathbb{F}_p^\times)^2$;
- (iii) whether $Q \in \langle P \rangle$, and if so the least nonnegative slope;
- (iv) a primitive-root discrete-log matrix B_p ;
- (v) the exact image size $|(\mathbb{Z}/(p-1)\mathbb{Z})^2 B_p|$ from the determinantal formula (26);
- (vi) whether $p-1 \mid \det B_p$, the rank-one-scale condition of Corollary 6.2.

The direct box enumeration and the Smith/determinantal image formula agreed for every row of the data set.

The four test pairs were:

$(8, 27),$	the exact integral pair $(2^3, 3^3)$;
$(5, 13),$	a generic integer control;
$(3779, 467744),$	an exceptionally close real-logarithm near-miss;
$(9, 2),$	a deliberately asymmetric control.

For the near-miss,

$$\begin{aligned} \log_2(3779) &= 11.883026394927862\dots, \\ \log_3(467744) &= 11.883026393570518\dots, \end{aligned} \tag{50}$$

whose discrepancy is only $1.3573444583858671 \times 10^{-9}$.

11.2 Results

Let “large defect” mean defect at least $m/4 = 144$. The quantity in the penultimate column below is the one-level row sum

$$\mathcal{E}_{\leq 5000}^{\text{row}} := \sum_{\substack{p \leq 5000 \\ p \nmid 6MA}} (m - c_p) \log p.$$

The integral pair is separated perfectly: every good prime is locally cyclic with the same slope 3, every discrete-log determinant vanishes modulo $p-1$, and every box collapses. This is the finite-field shadow of the global identity $Q = P^3$; the associated integer grid itself has repeated rows, so this control case is not an admissible positive determinant.

pair (M, A)	good p	collapse primes	large defect	local cyclic	rank-one scale	det = 0 mod $p - 1$	max defect	$\mathcal{E}$
(8, 27)	667	667	667	667	667	667	572	2,376,056.63
(5, 13)	665	13	11	1	2	1	540	12,425.85
(3779, 467744)	664	15	14	2	2	1	572	15,007.14
(9, 2)	667	14	11	0	0	0	552	13,122.57

Table 2: Exact one-level row-orbit statistics for $N = 24$ and good primes $p \leq 5000$. “Collapse” means $c_p < m$. “Local cyclic” means $\bar{Q}_p \in \langle \bar{P}_p \rangle$.

The three nonintegral controls look qualitatively alike. Almost every prime has the full 24×24 box image. Large collapses are rare, and local cyclicity is rarer still. In particular, the 10^{-9} agreement in (50) does not produce a corresponding broad modular signature. Among the exceptional primes:

- for (5, 13), $p = 37$ has local slope 30, image size 36, and defect 540; $p = 47$ has rank-one-scale image size 46 but no local slope;
- for the near-miss, $p = 5$ has local slope 2, image size 4, and defect 572; $p = 379$ has local slope 362, image size 189, and defect 387.

The second near-miss slope is large compared with the side length 24, so it gives no direct short translation inside the test box. This illustrates the difference between local cyclicity and useful local cyclicity.

11.3 What the experiment establishes and what it does not

The run verifies five implementation-level facts relevant to a future proof search:

- (1) the direct orbit count agrees exactly with the discrete-log Smith formula;
- (2) transpose duality predicts identical full-subgroup cardinalities, even though truncated row and column boxes may behave differently;
- (3) exact integral dependence creates a uniform, prime-by-prime signature;
- (4) extremely accurate real-logarithm matching does not by itself create that signature;
- (5) the informative exceptional primes often carry large slopes, confirming the need to record relation length rather than only image rank.

It does *not* test prime powers p^t , does not enumerate the column box, and does not approach the asymptotic regime required by (44). It also cannot rule out a sparse but eventually decisive family of primes beyond 5000.

11.4 Reproducibility record

The run produces `finite_field_scout.csv` and `finite_field_scout_summary.json`. The complete Python source is embedded in Appendix C; it uses only the Python standard library.

12 A formalization-ready decomposition

The finite-field route has an advantage over several analytic approaches in the source report: its central implications can be decomposed into small algebraic statements with explicit

certificates. This section records a Lean-oriented architecture. It is a design, not a claim that the formalization has already been completed.

12.1 Core objects

A formal development can avoid transcendental matrix entries until the final positivity lemma. Fix natural numbers M, A, U, V, R, S and define:

- (F1) finite index types for the row and column boxes;
- (F2) the integer entry function

$$E(a, b, r, s) = (2^a M^b)^r (3^a A^b)^s;$$

- (F3) the square integer matrix and its determinant D ;
- (F4) for each modulus q , the reduction maps ι_q and ι_q^* ;
- (F5) finite sets representing their box images.

All exponentiation in this layer is natural-number exponentiation. The only assumptions needed for modular inverses are explicit coprimality hypotheses.

12.2 The easiest certified divisor: one short relation

Theorem 5.2 admits a direct formal proof that does not require a general Smith-normal-form library:

- (L1) choose a spanning forest of the directed translation graph on the box;
- (L2) order vertices so that every non-root follows its parent;
- (L3) replace each non-root row by row minus parent row;
- (L4) prove that every entry in each replaced row is divisible by p^t ;
- (L5) factor p^t from each such row and use determinant invariance under row addition.

The number of non-root vertices is exactly the overlap count. This yields a compact checker whose input is only (p, t, h, k) plus two modular exponentiation equalities.

12.3 Tomography without choosing a Smith normal form

The full layer theorem can be formalized in either of two ways.

Smith route. Use the Smith normal form over $\mathbb{Z}$ and show that the image of the reduced diagonal matrix over $\mathbb{Z}/p^t\mathbb{Z}$ has minimal generator number $\#\{i : v_p(s_i) < t\}$. Bound that generator number by the number of distinct rows or columns.

Determinantal-ideal route. For each j , show that if the reduced row module is generated by at most c elements modulo p^t , then every $(c+1) \times (c+1)$ minor is divisible by p^t . Apply the relation between determinantal ideals and partial products of invariant factors, then sum over t . This route may align better with existing commutative-algebra infrastructure.

The short-relation certificate should be formalized first because it already captures the moving-slope target and has a smaller dependency surface.

12.4 The positivity boundary

To use a nonzero determinant, the formal development needs the fact that

$$\det(e^{x_i y_j})_{1 \leq i, j \leq m} > 0$$

for strictly increasing real sequences x_i and y_j . Several proof paths are available:

- (P1) invoke strict total positivity of the exponential kernel;
- (P2) prove the generalized Vandermonde determinant is positive by repeated integral factorization;
- (P3) use the Chebyshev-system criterion for the functions $e^{y_j x}$.

For this application, strict ordering follows from irrationality of β and irrationality of $\log_3 2$. Irrationality of β is elementary from the integrality assumptions: a rational exponent producing an integer power of 2 must be integral.

12.5 A machine-checkable terminal certificate

A finite successful certificate could have the following serialized form:

```
(M, A, U, V, R, S,
 [(p, t, row_image_partition, column_image_partition), ...],
 rational_upper_bound)
```

The checker would:

- (C1) verify primality and good reduction;
- (C2) recompute every modular row and column image;
- (C3) sum the exact Smith-layer lower bound;
- (C4) verify an outward-rounded rational upper bound for $mT_{U,V}L_{R,S} + \frac{m}{2} \log m$;
- (C5) compare the two rational numbers.

No trust in the search program would be required. The search can be heuristic; only the small certificate checker needs formal verification.

13 Research targets produced by the finite-field route

The route does not yet prove the conjecture. It does, however, replace a diffuse search by several sharply falsifiable statements. Each target below comes with a precise way it would close, or fail to close, the determinant argument.

13.1 Target I: moving-slope logarithmic mass

Let U_m, V_m be aspect-optimized row sides with $U_m V_m = m$, and for $0 < \eta < 1$ define

$$\mathcal{P}_m(\eta) := \{p \nmid 6MA : \exists c \in \mathbb{Z}, |c| \leq \eta U_m, Q \equiv P^c \pmod{p}\}.$$

A sufficient theorem is

$$\limsup_{m \rightarrow \infty} \frac{1}{m} \sum_{p \in \mathcal{P}_m(\eta)} \log p > \frac{4\sqrt{\beta \log 2 \log 3}}{1 - \eta}. \quad (51)$$

By (43), this would contradict a nonintegral pair. Theorem 7.4 shows that any proof must exploit a growing collection of slopes. A direct attack could study the prime divisors of the moving family

$$\gcd(M - 2^c, A - 3^c), \quad |c| \leq \eta U_m,$$

with multiplicity and with overlap between different c handled by the lcm rather than the product.

A falsifier would be a hypothetical pair for which the left side of (51) remains below the displayed threshold for every $\eta < 1$. Even such a negative result would be valuable because it would rule out the most direct one-relation determinant closure.

13.2 Target II: p -adic depth without short level-one slopes

Level-one local slopes may be sparse, but a prime can contribute through several Smith layers. Define the best short-slope depth

$$d_p(U) := \max_{|c| < U} \min\{v_p(M - 2^c), v_p(A - 3^c)\},$$

with the negative- c expressions cleared of denominators. The short-relation theorem gives roughly

$$(U - |c|)(V - 1)d_p(U) \log p$$

from the maximizing slope. A depth version of Target I is therefore

$$\sum_p d_p(U_m) (1 - |c_p|/U_m)_+ \log p > \left(4\sqrt{\beta \log 2 \log 3} + o(1)\right) m. \quad (52)$$

The challenge is simultaneous lifting in two coordinates. Ordinary Hensel lifting of a discrete logarithm in one coordinate does not force the same lifted slope in the other.

13.3 Target III: two independent short relations

One local relation collapses a box along one direction. If $L_{p^t}^{\text{row}}$ contains two independent vectors u_1, u_2 whose fundamental parallelogram has small area relative to UV , then the orbit count can be much smaller than any one-overlap estimate. Geometry of numbers suggests studying the successive minima

$$\lambda_1(L_{p^t}^{\text{row}}), \quad \lambda_2(L_{p^t}^{\text{row}})$$

with respect to the rectangle norm $\|(h, k)\|_{U,V} = \max(|h|/U, |k|/V)$. The exact theorem needed is a lower bound on

$$\sum_{p,t} \left(m - \#(\Omega_{U,V}/L_{p^t}^{\text{row}})\right) \log p$$

that exploits both minima, not only the lattice index. Discrete-log determinants control the index; the unresolved information is the shape.

13.4 Target IV: sharpen the archimedean determinant

Hadamard discards all cancellation and the precise total-positive structure. For increasing x_i, y_j there are exact factorizations

$$\det(e^{x_i y_j}) = e^{\sum_i x_i y_i} \prod_{i < j} (x_j - x_i) \prod_{i < j} (y_j - y_i) \cdot \mathcal{I}(x, y),$$

where $\mathcal{I}$ is a positive iterated-integral factor. A uniform upper bound that saves a fixed proportion of the leading constant in (40) would immediately lower the required local mass. To be decisive against the cyclic-cardinality ceiling alone, the leading constant would have to fall below $1/2$, an enormous improvement; more plausibly, a moderate saving would combine with short-relation divisors.

13.5 Target V: transfer singularity of logarithms to finite places

The defining real relation is the rank-one identity

$$\det \begin{pmatrix} \log 2 & \log M \\ \log 3 & \log A \end{pmatrix} = 0.$$

At a prime with chosen primitive root, the finite analogue is

$$\det B_p \equiv 0 \pmod{p-1}.$$

A theorem forcing this congruence for a positive logarithmic mass of primes would already yield many rank-one-scale images, but Proposition 8.3 shows that image size alone is not enough. The transfer theorem must be refined to produce small kernel vectors or substantial p -adic depth. This is the conceptual bottleneck:

convert an exact singularity in the real logarithm matrix into quantitative shape information for the kernels of its finite logarithm matrices.

No known support theorem provides such a transfer.

14 Conclusions

The investigation did not produce a complete proof of the Alaoglu–Erdős conjecture. It produced a new finite-place framework with exact theorems, exact computational diagnostics, and several negative results that materially narrow the remaining search.

The principal advances are:

- (1) *Pair-specific Smith tomography.* The exact reductions of $(2, 3)$ and (M, A) modulo every p^t bound every Smith layer of the integer grid determinant; this is stronger and more tailored than a universal polynomial-function divisor.
- (2) *Short-relation amplification.* A local relation (h, k) contributes its geometric overlap multiplicity, yielding explicit weighted row and column gcd divisors.
- (3) *Finite closure.* The orbit-collapse energy (35) gives a completely finite contradiction certificate against the Hadamard budget (34).

- (4) *Sharp route diagnostics.* Cyclic image size at every prime-power level contributes only $(1/2 + o(1))m^2$ under the present estimate; every fixed slope and every finite family of fixed relation rays are also insufficient at the leading m^2 scale.
- (5) *Local–global boundary.* Existing support theorems would prove integrality from almost-everywhere local cyclicity or order domination, but a counterexample must have infinitely many anti-collapse witness primes. What is missing is quantitative control of the complementary short-relation mass.
- (6) *Reproducible reconnaissance.* Exact finite-field computations sharply distinguish a true integral-power pair from generic pairs and from an archimedean near-miss, while confirming the discrete-log image formula on every tested prime.

The strongest concrete next statement is (51), or its depth and two-relation variants. A proof of one of these quantitative targets, together with Theorem 8.2, would finish the conjecture. Conversely, a systematic failure would establish that the determinant route requires a fundamentally sharper archimedean ingredient. The problem has therefore been reduced, within this approach, to a precise local-shape question rather than another equivalent reformulation of four exponentials.

A Expanded proof of the Smith-layer identity

For completeness, this appendix records the module-theoretic identity used in Theorem 4.2. Let $E \in M_m(\mathbb{Z})$ be nonsingular and write its Smith form over $\mathbb{Z}_p$ as

$$UEV = \text{diag}(p^{\nu_1}, \dots, p^{\nu_m}), \quad \nu_1 \leq \dots \leq \nu_m,$$

with $U, V \in \text{GL}_m(\mathbb{Z}_p)$. Reduction modulo p^t preserves the isomorphism class of the row and column modules. The row module of the diagonal matrix over $R_t = \mathbb{Z}/p^t\mathbb{Z}$ is

$$\bigoplus_{\nu_i < t} p^{\nu_i} R_t.$$

Each summand is isomorphic to $R_t/p^{t-\nu_i}R_t$ and contributes one dimension to M/pM over $\mathbb{F}_p$. Nakayama’s lemma therefore gives

$$\mu_{R_t}(\text{row}(E \bmod p^t)) = \#\{i : \nu_i < t\} = r_{p,t},$$

where μ denotes minimal generator number. If only c distinct rows occur, those rows generate the module, so $r_{p,t} \leq c$. The same argument applies to columns.

Finally, for every nonnegative integer ν ,

$$\nu = \sum_{t \geq 1} \mathbf{1}_{\nu \geq t}.$$

Summing over the Smith valuations gives

$$v_p(\det E) = \sum_i \nu_i = \sum_{t \geq 1} \#\{i : \nu_i \geq t\} = \sum_{t \geq 1} (m - r_{p,t}).$$

This proves the layer-cake identity and explains why counting image classes at *all* prime-power depths is exactly matched to determinant valuation.

B A direct elimination proof of the short-relation divisor

Let $\Omega \subset \mathbb{Z}^2$ be finite and let $u \neq 0$ satisfy $P^{u_1}Q^{u_2} \equiv 1 \pmod{p^t}$. Form a graph on Ω by joining ω and $\omega + u$ whenever both belong to Ω . Because translation by a nonzero vector has no finite cycles in $\mathbb{Z}^2$, this graph is a forest. Choose one root in each component and orient every edge away from the root.

Order the rows so that parents precede children. For every non-root vertex ω , replace the row indexed by ω by

$$\text{row}(\omega) - \text{row}(\text{parent}(\omega)).$$

The operation preserves the determinant. The two row parameters differ by u , so every entry in the new row is divisible by p^t . Factoring p^t from every non-root row gives

$$p^{t(|\Omega| - \#\text{components})} \mid D.$$

A forest has $|\Omega| - \#\text{components}$ edges. For a rectangle and a single translation $u = (h, k)$, that edge count is the exact overlap $(U - |h|)_+(V - |k|)_+$. This proof is especially convenient for formal verification because it avoids any appeal to Smith normal form.

C Exact reconnaissance code

The following is the complete standard-library Python program used for Table 2. It writes both the row-level CSV file and the JSON summary.

```

1  #!/usr/bin/env python3
2  """Finite-field orbit scout for the Alaoglu--Erdős grid.
3
4  For each good prime p, the program computes:
5  * the discrete-log matrix B_p for P=(2,3), Q=(M,A),
6  * the full orbit size |im B_p| in (Z/(p-1)Z)^2,
7  * exact row/column image counts for the N x N exponent box,
8  * whether Q mod p lies in the cyclic subgroup generated by P mod p.
9
10 All arithmetic is exact integer/modular arithmetic. This is exploratory evidence,
11 not a proof of the conjecture.
12 """
13 from __future__ import annotations
14
15 import argparse
16 import csv
17 import json
18 import math
19 from pathlib import Path
20 from typing import Dict, Iterable, List, Sequence, Tuple
21
22 Pair = Tuple[int, int, str]
23
24
25 def primes_upto(limit: int) -> List[int]:
26     if limit < 2:
27         return []
28     sieve = bytearray(b"\x01") * (limit + 1)
29     sieve[0:2] = b"\x00\x00"
30     stop = int(limit**0.5)
31     for q in range(2, stop + 1):
32         if sieve[q]:
33             start = q * q
34             sieve[start : limit + 1 : q] = b"\x00" * ((limit - start) // q + 1)
35     return [q for q in range(2, limit + 1) if sieve[q]]
36
37
38 def prime_factors(n: int) -> List[int]:
39     ans: List[int] = []
40     d = 2
41     while d * d <= n:
42         if n % d == 0:
43             ans.append(d)
44             while n % d == 0:
45                 n //= d
46     d = 3 if d == 2 else d + 2
47     if n > 1:
48         ans.append(n)
49     return ans
50
51

```

```

52 def primitive_root_prime(p: int) -> int:
53     if p == 2:
54         return 1
55     n = p - 1
56     factors = prime_factors(n)
57     for g in range(2, p):
58         if all(pow(g, n // ell, p) != 1 for ell in factors):
59             return g
60     raise RuntimeError(f"No primitive root found modulo {p}")
61
62
63 def discrete_log_table(p: int, g: int) -> Dict[int, int]:
64     table: Dict[int, int] = {}
65     x = 1
66     for k in range(p - 1):
67         table[x] = k
68         x = (x * g) % p
69     if len(table) != p - 1:
70         raise RuntimeError(f"{g} is not primitive modulo {p}")
71     return table
72
73
74 def image_size_mod_n(matrix: Tuple[int, int, int, int], n: int) -> int:
75     """Cardinality of the image of a 2x2 integer matrix on  $(\mathbb{Z}/n\mathbb{Z})^2$ .
76
77     If  $B=(b_{ij})$ , the cokernel has order equal to the gcd of the 2x2 minors
78     of  $[n \ I_2 \mid B]$ .
79     """
80     b11, b12, b21, b22 = matrix
81     det = b11 * b22 - b12 * b21
82     cokernel = math.gcd(
83         n * n,
84         n * b11,
85         n * b12,
86         n * b21,
87         n * b22,
88         abs(det),
89     )
90     if cokernel <= 0 or (n * n) % cokernel:
91         raise RuntimeError("Invalid lattice index")
92     return (n * n) // cokernel
93
94
95 def powers(base: int, length: int, modulus: int) -> List[int]:
96     vals = [1]
97     b = base % modulus
98     for _ in range(1, length):
99         vals.append((vals[-1] * b) % modulus)
100     return vals
101
102
103 def box_image_counts(M: int, A: int, p: int, N: int) -> Tuple[int, int]:
104     p2 = powers(2, N, p)
105     p3 = powers(3, N, p)
106     pM = powers(M, N, p)
107     pA = powers(A, N, p)
108
109     row_points = {
110         ((p2[a] * pM[b]) % p, (p3[a] * pA[b]) % p)
111         for a in range(N)
112         for b in range(N)
113     }
114     column_signatures = {
115         ((p2[r] * p3[s]) % p, (pM[r] * pA[s]) % p)
116         for r in range(N)
117         for s in range(N)
118     }
119     return len(row_points), len(column_signatures)
120
121
122 def local_cyclic_slope(M: int, A: int, p: int) -> Tuple[bool, int | None, int]:
123     """Return whether  $Q$  is in  $\langle P \rangle$ , a least nonnegative slope, and  $\text{ord}(P)$ ."""
124     target = (M % p, A % p)
125     x, y = 1, 1
126     slope: int | None = None
127     for k in range(p - 1):
128         if slope is None and (x, y) == target:
129             slope = k
130             x = (x * 2) % p
131             y = (y * 3) % p
132             if (x, y) == (1, 1):
133                 order = k + 1
134             return slope is not None, slope, order
135     raise RuntimeError(f"Failed to find the order of P modulo {p}")
136
137
138 def slope_mismatch(M: int, A: int) -> float:
139     return math.log(M) / math.log(2.0) - math.log(A) / math.log(3.0)
140
141
142 def analyze_pair(M: int, A: int, label: str, N: int, prime_limit: int) -> Tuple[List[dict], dict]:
143     if M <= 0 or A <= 0 or N <= 0:
144         raise ValueError("M, A, and N must be positive")
145
146     rows: List[dict] = []
147     R = N * N
148     weighted_defect = 0.0

```

```

149 primes_with_defect = 0
150 primes_with_large_defect = 0
151 cyclic_primes = 0
152 rank_one_scale_primes = 0
153 determinant_zero_primes = 0
154 max_defect = 0
155
156 for p in primes_upto(prime_limit):
157     if (6 * M * A) % p == 0:
158         continue
159     g = primitive_root_prime(p)
160     logs = discrete_log_table(p, g)
161     matrix = (
162         logs[2 % p],
163         logs[M % p],
164         logs[3 % p],
165         logs[A % p],
166     )
167     det_integer = matrix[0] * matrix[3] - matrix[1] * matrix[2]
168     det_mod = det_integer % (p - 1)
169     full_image = image_size_mod_n(matrix, p - 1)
170     row_count, col_count = box_image_counts(M, A, p, N)
171     defect = max(R - row_count, R - col_count)
172     cyclic, slope, orderP = local_cyclic_slope(M, A, p)
173
174     weighted_defect += defect * math.log(p)
175     primes_with_defect += int(defect > 0)
176     primes_with_large_defect += int(defect >= R / 4)
177     cyclic_primes += int(cyclic)
178     rank_one_scale_primes += int(full_image <= p - 1)
179     determinant_zero_primes += int(det_mod == 0)
180     max_defect = max(max_defect, defect)
181
182     rows.append(
183         {
184             "label": label,
185             "M": M,
186             "A": A,
187             "p": p,
188             "primitive_root": g,
189             "b11": matrix[0],
190             "b12": matrix[1],
191             "b21": matrix[2],
192             "b22": matrix[3],
193             "det_mod_p_minus_1": det_mod,
194             "full_orbit_image_size": full_image,
195             "row_box_image_size": row_count,
196             "column_box_image_size": col_count,
197             "box_rank_defect_lower_bound": defect,
198             "Q_in_cyclic_subgroup_P": int(cyclic),
199             "least_nonnegative_local_slope": "" if slope is None else slope,
200             "order_of_P": orderP,
201         }
202     )
203
204     summary = {
205         "label": label,
206         "M": M,
207         "A": A,
208         "box_side_N": N,
209         "box_cardinality": R,
210         "prime_limit": prime_limit,
211         "slope_mismatch_log2M_minus_log3A": slope_mismatch(M, A),
212         "good_primes": len(rows),
213         "primes_with_box_collision": primes_with_defect,
214         "primes_with_defect_at_least_one_quarter": primes_with_large_defect,
215         "primes_with_local_cyclicity": cyclic_primes,
216         "primes_with_rank_one_scale_full_image": rank_one_scale_primes,
217         "primes_with_discrete_log_determinant_zero": determinant_zero_primes,
218         "maximum_box_defect_lower_bound": max_defect,
219         "sum_defect_times_log_p": weighted_defect,
220     }
221     return rows, summary
222
223
224 def parse_pairs(specs: Sequence[str]) -> List[Pair]:
225     pairs: List[Pair] = []
226     for spec in specs:
227         parts = spec.split(":")
228         if len(parts) not in (2, 3):
229             raise ValueError(f"Bad pair specification: {spec}")
230         M, A = int(parts[0]), int(parts[1])
231         label = parts[2] if len(parts) == 3 else f"M={M},A={A}"
232         pairs.append((M, A, label))
233     return pairs
234
235
236 def main() -> None:
237     parser = argparse.ArgumentParser()
238     parser.add_argument("--N", type=int, default=24)
239     parser.add_argument("--prime-limit", type=int, default=5000)
240     parser.add_argument(
241         "--pair",
242         action="append",
243         default=[],
244         help="M:A[:label]; may be repeated",
245     )

```

```

246 parser.add_argument("--csv", type=Path, required=True)
247 parser.add_argument("--summary", type=Path, required=True)
248 args = parser.parse_args()
249
250 pairs = parse_pairs(args.pair) if args.pair else [
251     (8, 27, "integral slope beta=3"),
252     (5, 13, "small generic pair"),
253     (3779, 467744, "extreme archimedean near-miss"),
254     (9, 2, "synthetic coprime pair"),
255 ]
256
257 all_rows: List[dict] = []
258 summaries: List[dict] = []
259 for M, A, label in pairs:
260     rows, summary = analyze_pair(M, A, label, args.N, args.prime_limit)
261     all_rows.extend(rows)
262     summaries.append(summary)
263
264 args.csv.parent.mkdir(parents=True, exist_ok=True)
265 with args.csv.open("w", newline="", encoding="utf-8") as fh:
266     writer = csv.DictWriter(fh, fieldnames=list(all_rows[0].keys()))
267     writer.writeheader()
268     writer.writerows(all_rows)
269
270 payload = {
271     "summaries": summaries,
272 }
273 args.summary.write_text(json.dumps(payload, indent=2, sort_keys=True) + "\n", encoding="utf-8")
274 print(json.dumps(payload, indent=2, sort_keys=True))
275
276
277 if __name__ == "__main__":
278     main()

```

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